Manas Kshirsagar

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WORK EXPERIENCE

AI & Automation Intern, at Atlas Copco Group **Aug 2026 – Present**

- Building an AI-powered resume ranking pipeline using NLP and ML-based scoring to automate candidate screening workflows
- Developing a machine learning driven sales lead pipeline integrating predictive analytics for refined lead scoring and campaign recommendation
- Implementing a Databricks based monitoring and inference solution enabling mechanical engineers to analyze operational data in real time

Assistant Researcher, at Vishwakarma Institute of Technology, Pune, MH **Mar 2026 – Present**

- Secured NVIDIA Academic Grant Program support (32,000 A100 GPU-hours) through a self-authored technical research proposal, enabling large-scale scientific computing infrastructure for multi-band JWST astronomical image processing
- Developed high-performance C++ data loading modules for large-scale scientific computing workflows, processing multi-filter astronomical imagery (NIRCam + MIRI, 8-17 spectral bands) with low-level memory optimization and efficient host-device data transfer on NVIDIA A100 hardware
- Designed and profiled GPU-accelerated inference pipelines using CUDA memory hierarchies and kernel execution optimization, applying NVIDIA Nsight to identify and resolve bottlenecks in vision transformer training and inference code paths across distributed scientific computing workloads.

TECHNICAL SKILLS

Languages: Python, C++, Java, Go, SQL, JavaScript, Kotlin
Machine Learning: Neural Networks, Vision Transformers, Probabilistic Modeling, Natural Language Processing,
Generative AI & LLMs: MCP, RAG, FAISS, LangChain, Prompt Engineering, Vector Search, Diffusion Models, Nemotron
DevOps & Cloud: Docker, Kubernetes, AWS (EC2, S3, API Gateway), FastAPI, Grafana, OpenTelemetry, Git, Linux, CI/CD
Data Engineering: Databricks, PostgreSQL, Numpy, xarray, Parquet, MySQL, NoSQL
Miscellaneous: PyTorch, API Engineering, CUDA, NVIDIA Nsight, Astropy, SciPy, Time Series Analysis

PROJECTS

Derivative Hedging using Deep Learning & RL | PPO-based Options Strategy **(Jan 2026 – Apr 2026)**

Tech: Python, PyTorch, PPO, SciPy, Gymnasium, custom RL environments

- Applied probabilistic modelling and statistical inference to build a PPO-based reinforcement learning system for options hedging across a 12-instrument portfolio, achieving 96.8% reduction in hedging error versus Black-Scholes baseline ($p<0.05$)
- Iteratively resolved degenerate training behaviours (momentum exploits, position gaming) through reward engineering, curriculum learning, and action masking, demonstrating systematic debugging and mathematical validation methodology
- Designed validation pipelines for unseen market data, maintaining 86.3% variance reduction across out-of-sample instruments.

FloatChat | Observability & Inference for Oceanic Data **(Sep 2025 – Dec 2025)**

Tech: Python, LangChain, FAISS, RAG, Grafana, OpenTelemetry, AWS

- Developed a full-stack scientific data platform for processing and analyzing large-scale oceanographic datasets (70+ features, 50K+ instances), implementing ETL pipelines, distributed inference workflows, and real-time observability infrastructure
- Designed ETL pipelines and data-driven predictive analytics integrations across structured and unstructured datasets (70+ features, 50K+ instances), with full documentation of system architecture and workflows.

More: https://github.com/MaxVon89?tab=repositories

CERTIFICATIONS

NVIDIA Deep Learning Institute | 2 Certifications: Anomaly Detection with AI **&** Gen AI with Diffusion Models
Completed advanced DLI modules covering production-grade deep learning techniques used in industry research and deployment.

IBM:
DevOps and Software Engineering Professional Certificate (Coursera)
15-course program covering CI/CD, containerization (Docker, Kubernetes, OpenShift), microservices, Agile methodologies, cybersecurity, and secure cloud-native application deployment.

EDUCATION

B. Tech (Artificial Intelligence & Data Science), at Vishwakarma Institute of Tech. **2023 –**
Grade (GPA): 8.92/10
Relevant Coursework: Artificial Intelligence, Machine Learning, Deep Learning, Linear Algebra, Discrete Mathematics, Theory of Computation, Statistical Inference, Data Modelling

EXTRACURRICULAR

Lead, at Microsoft Learn Student Club: **2024 – 2026**

Led a 6-member multimedia team to produce branding content reaching 3,000+ students across campus events.
Secured industry sponsorships and coordinated a workshop with a Google AI consultant, increasing event participation by 40% and coordinating with international industry professionals across technical and non-technical domains.
Maintained documentation for team workflows, sponsorship processes, and event planning across 2 years.